

Machine Learning — Programming Assignment

September 2026

1 Problem definition and data

Domotics offers a wide range of fascinating problems that machine learning tools can address. A notable example is occupancy detection: determining whether a room has people inside without relying on privacy-invasive sensors, such as video cameras.

A dataset has been collected for this purpose. It includes 8143 records reporting measures from sensors in a room. Each record includes the following information:

1. Temperature, in Celsius degrees;
2. Humidity, in percentage;
3. Light, in Lux;
4. CO₂, in ppm;
5. Humidity ratio, calculated using the measured temperature and relative humidity;
6. Occupancy: indicating the presence or absence of people.

The last feature is binary, and is the one we want to predict, given the others.

The dataset has already been divided into a training set of 6943 records, a validation set of 400 records, and a test set of 800 records. Each set is stored as a text file, with one record for each row, and with the features separated by spaces.

2 Assignment

We want to build one or more classifiers that automatically determine whether or not people are in the monitored room.

For the programming assignment, you are expected to:

1. analyze and comment the data;

2. implement, train, and evaluate one or more classification models;
3. use suitable data processing and visualization techniques to analyze the behavior of the trained models.

All the above should be implemented as scripts in the Python programming language. The code written during the course can be used for the assignment if needed.

Special request: we want multiple version of the model (or the models). We want at least a *precision-oriented* version, a *recall-oriented* version, and one or more intermediate versions. Performance should be visualized using a ROC (precision/recall) curve.

3 Report

Prepare a report of three to five pages documenting all your work. Provide detailed instructions on how to reproduce the results. The report must be in the PDF format. Include your name in the report and conclude the document with the following statement: “I affirm that this report is the result of my own work and that I did not share any part of it with anyone else except the teacher.”

Make a ZIP archive with the report and the Python notebook (converted to PDF, with visible outputs), and submit it from the course web page.